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Marks	10.78/13.00
Grade	8.29 out of 10.00 (82.95%)

Question 1

Correct

Mark 1.00 out of 1.00

What is the goal of Imitation Learning?

Select one or more:

- ☐ a. Given a collection of episodes generated by an expert, or the expert itself, train a policy to maximize the future expected reward
- ☒ b. Given a collection of episodes generated by an expert, or the expert itself, train a policy to produce the expert demonstrations ✓
- ☐ c. Given an environment, a policy class and a loss function, train a policy to mimic the optimal policy
- ☐ d. Given a collection of demonstrations or a demonstrator, train a policy to be equal or better than the demonstrator in terms of average score.

Your answer is correct.

The correct answer is: Given a collection of episodes generated by an expert, or the expert itself, train a policy to produce the expert demonstrations

Question 2

Correct

Mark 1.00 out of 1.00

Assume that the learned policy π_θ makes a mistake with respect to the expert policy $\check{\pi}$ with probability ϵ in every state.

Why the compounding errors grow quadratically in T ?

Select one or more:

- ☐ a. This is only true when the cost is defined as the mean square error.
- ☒ b. Because when the learned policy π_θ makes a mistake it can end up in states that the expert policy $\check{\pi}$ never visited before and therefore incur from that point the maximal cost. ✓
- ☐ c. Because when the expert policy $\check{\pi}$ makes a mistake it can end up in states that the learned policy π_θ never visited before and therefore incur from that point the maximal cost.
- ☐ d. This is only true when the loss function is the mean square error.

Your answer is correct.

The correct answer is: Because when the learned policy π_θ makes a mistake it can end up in states that the expert policy $\check{\pi}$ never visited before and therefore incur from that point the maximal cost.

Question 3

Correct

Mark 1.00 out of 1.00

What is/are the limitations of Behavioral Cloning?

Select one or more:

- ☐ a. It is limited only to function approximation (i.e. Neural Networks), where generalization over states is possible.
- ☒ b. You will have to deal with distribution shift produced by accumulation of compounding errors ✓
- ☐ c. It is limited to the tabular case, where every state has its own predictor.
- ☒ d. You need the states and the expert's actions along the trajectory. ✓
- ☐ e. It is limited to Markov Decision Processes (MDPs), where all the information you need to predict the action is encoded in the current state.

Your answer is correct.

The correct answers are: You need the states and the expert's actions along the trajectory., You will have to deal with distribution shift produced by accumulation of compounding errors

Question 4

Correct

Mark 1.00 out of 1.00

Why does causal confusion happen in Imitation Learning?

Select one or more:

- ☐ a. Causal confusion only happens when using Behavioral Cloning in Partial Observable Markov Decision Processes (POMDPs)
- ☐ b. Causal confusion only happens in Behavioral Cloning
- ☒ c. Because the training procedure (Supervised Learning) is unaware of the causal structure on the interaction between the policy and the environment. ✓
- ☐ d. Because the expert is not always selecting the same action in a given state (i.e. irrelevant actions, different strategies, demonstrations from different experts)
- ☒ e. Because Supervised Learning learns correlations which does not necessarily imply causation. ✓

Your answer is correct.

The correct answers are: Because the training procedure (Supervised Learning) is unaware of the causal structure on the interaction between the policy and the environment., Because Supervised Learning learns correlations which does not necessarily imply causation.

Question 5

Correct

Mark 1.00 out of 1.00

Under what circumstances, Interactive Expert methods (as studied in class) are possible?

Select one or more:

- ☐ a. When the state and action space is small enough to use a table (tabular case)
- ☐ b. When we have the model of the environment $p(s' | s, a)$
- ☐ c. When we have enough data to use Neural Networks
- ☒ d. When we have an expert that we can ask ✓

Your answer is correct.

The correct answer is: When we have an expert that we can ask

Question 6

Correct

Mark 1.00 out of 1.00

What are the benefits of Imitation Learning in Reinforcement Learning (in general terms)?

Select one or more:

- ☐ a. It gives you the optimal policy almost for free
- ☐ b. It is an valid alternative to get the optimal policy
- ☒ c. It helps with exploration in high dimensional state and action spaces ✓
- ☒ d. It gives you a good initialization of your agent ✓
- ☐ e. It is a powerful way to overcome catastrophic forgetting
- ☒ f. It is very easy to implement ✓
- ☒ g. It is very useful when rewards are sparse ✓

Your answer is correct.

The correct answers are: It is very easy to implement, It helps with exploration in high dimensional state and action spaces, It gives you a good initialization of your agent, It is very useful when rewards are sparse

Question 7

Partially correct

Mark 0.42 out of 1.00

What were the problems that ALVINN faced back in the late 80's?

Select one or more:

- ☒ a. Different distributions in training and test data ✓
- ☐ b. Back in this time there was not computational power enough to process data "on-the-fly"
- ☒ c. The network was so small that it was forgetting past non i.i.d data (catastrophic forgetting) ✗
- ☐ d. Same distributions in training and test data
- ☐ e. Not enough variance in the training samples
- ☐ f. Too much variance in the training samples
- ☒ g. When training "on-the-fly" the data was not i.i.d and led to catastrophic forgetting ✓

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: Not enough variance in the training samples, Different distributions in training and test data, When training "on-the-fly" the data was not i.i.d and led to catastrophic forgetting

Question 8

Partially correct

Mark 0.67 out of 1.00

How Pomerleau's team mitigated the problems they faced when training a network to drive a car in the late 80's?

Select one or more:

- ☐ a. They created a replay buffer to store samples
- ☒ b. They store good and bad trajectories to keep the variance in the buffer high ✓
- ☐ c. They created a replay buffer to store low error samples
- ☐ d. They store good and bad trajectories to keep the variance in the buffer low
- ☐ e. They created a replay buffer to store high error samples
- ☒ f. They implement a sampling strategy to compensate underrepresented states or behaviours ✓
- ☐ g. They store only good trajectories to mimic the best policy
- ☐ h. They implement a sampling strategy to overfit on good behaviours (better policies)
- ☐ i. They store only bad trajectories to avoid bad policies

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: They created a replay buffer to store samples, They store good and bad trajectories to keep the variance in the buffer high, They implement a sampling strategy to compensate underrepresented states or behaviours

Question 9

Correct

Mark 1.00 out of 1.00

What of the following statements are TRUE regarding the connection between Imitation Learning (IL) and Inverse Reinforcement Learning (IRL)

Select one or more:

- ☐ a. IL and IRL are closely related since both learn good policies
- ☐ b. IL and IRL are closely related since both recover the reward function
- ☒ c. IL and IRL are closely related since both learn from demonstrations ✓
- ☐ d. IL and IRL are closely related since both learn a policy

Your answer is correct.

The correct answer is: IL and IRL are closely related since both learn from demonstrations

Question 10

Partially correct

Mark 0.50 out of 1.00

Given:

1. $\mathcal{L}(\cdot)$ is some loss function
2. $\breve{\pi}$ is the expert policy
3. $\mathcal{D}$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by $\breve{\pi}$
4. $p_{\text{data}}(s)$ is the distribution of states in $\mathcal{D}$
5. π_θ is the policy parametrized by θ
6. $\mathcal{D}_\theta$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π_θ
7. $p_{\pi_\theta}(s)$ is the distribution of states in $\mathcal{D}_\theta$

Which of the following expressions correspond to the learning objective of **Behavioral Cloning**?

Select one or more:

- ☐ a. $\operatorname{argmax}_{\theta} \sum_{s \in \mathcal{D}} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- ☐ b. $\operatorname{argmax}_{\theta} \mathbb{E}_{s \sim p_{\text{data}}} [\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))]$
- ☐ c. $\operatorname{argmax}_{\theta} \mathbb{E}_{s \sim p_{\breve{\pi}}} [\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))]$
- ☐ d. $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- ☐ e. $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$
- ☐ f. $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\breve{\pi}}} [\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))]$
- ☒ g. $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\text{data}}} [\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))]$ ✓
- ☐ h. $\operatorname{argmax}_{\theta} \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\text{data}}} [\mathcal{L}(\breve{\pi}(s), \pi_\theta(s))]$, $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}} \mathcal{L}(\breve{\pi}(s_i), \pi_\theta(s_i))$

Question 11

Partially correct

Mark 0.50 out of 1.00

Given:

1. $\mathcal{L}(\cdot)$ is some loss function
2. π^* is the expert policy
3. $\mathcal{D}$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π^*
4. $p_{\text{data}}(s)$ is the distribution of states in $\mathcal{D}$
5. π_θ is the policy parametrized by θ
6. $\mathcal{D}_\theta$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π_θ
7. $p_{\pi_\theta}(s)$ is the distribution of states in $\mathcal{D}_\theta$

Which of the following expressions correspond to the learning objective of **General Imitation Learning**?

Select one or more:

- ☒ a. $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\pi_\theta}} [\mathcal{L}(\pi^*(s), \pi_\theta(s))]$ ✓
- ☐ b. $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- ☐ c. $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- ☐ d. $\operatorname{argmax}_{\theta} \sum_{s \in \mathcal{D}} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- ☐ e. $\operatorname{argmax}_{\theta} \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$
- ☐ f. $\operatorname{argmax}_{\theta} \mathbb{E}_{s \sim p_{\text{data}}} [\mathcal{L}(\pi^*(s), \pi_\theta(s))]$
- ☐ g. $\operatorname{argmax}_{\theta} \mathbb{E}_{s \sim p_{\pi_\theta}} [\mathcal{L}(\pi^*(s), \pi_\theta(s))]$
- ☐ h. $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\text{data}}} [\mathcal{L}(\pi^*(s), \pi_\theta(s))]$

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: $\operatorname{argmin}_{\theta} \mathbb{E}_{s \sim p_{\pi_\theta}} [\mathcal{L}(\pi^*(s), \pi_\theta(s))]$

, $\operatorname{argmin}_{\theta} \sum_{s \in \mathcal{D}_\theta} \mathcal{L}(\pi^*(s_i), \pi_\theta(s_i))$

Question 12

Partially correct

Mark 0.90 out of 1.00

Given:

1. $\check{\pi}$ is the expert policy
2. $\mathcal{D}$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by $\check{\pi}$
3. π_θ is the policy parametrized by θ
4. $\mathcal{D}_\theta$ is the dataset of trajectories $\tau_j = \{(s_i, a_i)_{0:T}\}$ produced by π_θ

What is/are the correct implementation of the cross-entropy loss function for discrete actions in Behavioral Cloning?

Select one or more:

- ☐ a. $\mathcal{L}_{\text{BC}}(\mathcal{D}_\theta, \theta) := -\frac{1}{N} \sum_{k=1}^N (\pi_\theta(s_k) - a_k)^2$
- ☐ b. $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := -\sum_{k=1}^N \log \pi_\theta(a_k | s_k)$
- ☐ c. $\mathcal{L}_{\text{BC}}(\mathcal{D}_\theta, \theta) := -\sum_{k=1}^N (\pi_\theta(s_k) - a_k)^2$
- ☐ d. $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := \frac{1}{N} \sum_{k=1}^N \log \pi_\theta(a_k | s_k)$
- ☐ e. $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := \frac{1}{N} \sum_{k=1}^N (\pi_\theta(s_k) - a_k)^2$
- ☐ f. $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := \sum_{k=1}^N (\pi_\theta(s_k) - a_k)^2$
- ☐ g. $\mathcal{L}_{\text{BC}}(\mathcal{D}_\theta, \theta) := \sum_{k=1}^N \log \pi_\theta(a_k | s_k)$
- ☒ h. $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := -\frac{1}{N} \sum_{k=1}^N \log \pi_\theta(a_k | s_k)$ ✓

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := -\frac{1}{N} \sum_{k=1}^N \log \pi_\theta(a_k | s_k)$, $\mathcal{L}_{\text{BC}}(\mathcal{D}, \theta) := -\sum_{k=1}^N \log \pi_\theta(a_k | s_k)$ **Question 13**

Partially correct

Mark 0.80 out of 1.00

Supervised Learning works under the assumption that both **training** and **test** data are independent and identically distributed (**i.i.d.**). What is the reason or reasons why this assumptions does not hold in Imitation Learning (IL)?

Select one or more:

- ☐ a. Because in IL, both **training and test data are not i.i.d.** since future states depend on current actions
- ☒ b. Because in IL, **test data is not i.i.d.** since future states depend on current actions ✓
- ☐ c. Because in IL, **training data is not i.i.d.** due to the Markov property
- ☐ d. Because in IL, both **training and test data are not i.i.d.** due to the Markov property
- ☒ e. Because in IL, **test data is not i.i.d.** due to the Markov property ✗
- ☐ f. Because in IL, **training data is not i.i.d.** since future states depend on current actions

Your answer is partially correct.

You have selected too many options.

The correct answer is: Because in IL, **test data is not i.i.d.** since future states depend on current actions

